Lecture Unit 3.5

Accuracy

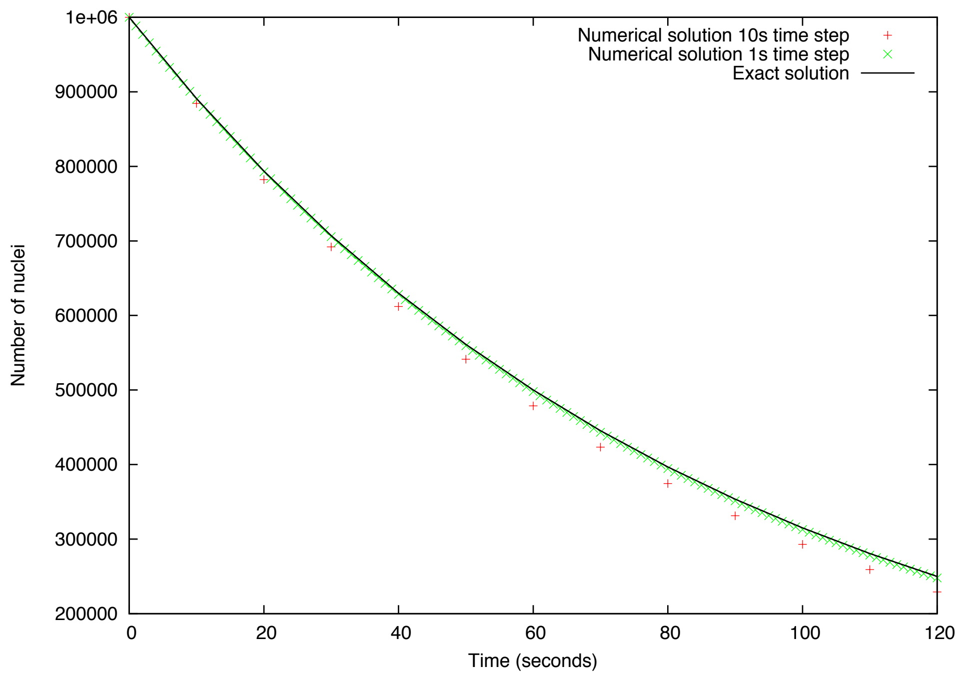
Exponential decay equation

- The numerical solution to the exponential decay equation was an approximation which showed divergence from the analytical solution

- This is because we assumed the rate of change was constant over the time step Δt

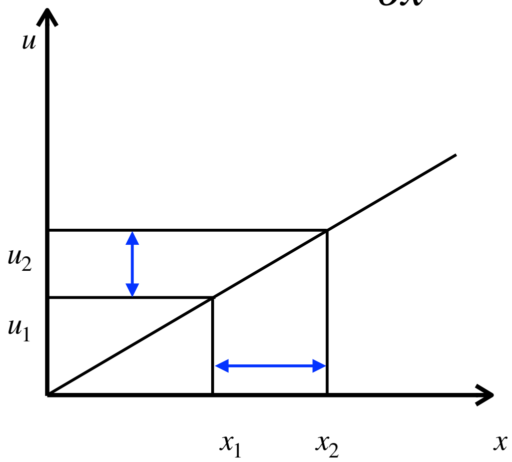
- We made the numerical solution more accurate by making the time step smaller

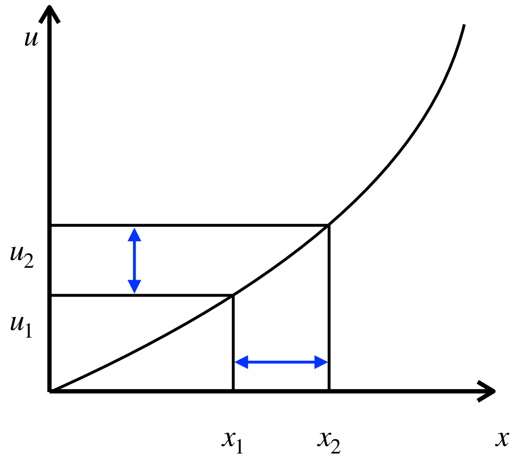
- Do finite differences for spatial derivatives have similar accuracy considerations?



Exponential decay equation

$$\frac{\partial u}{\partial x} \approx \frac{u_2 - u_1}{x_2 - x_1}$$





- If $u(x)$ is a straight line the gradient is the same everywhere and the approximation is exact

- If the line is curved the gradient changes and our approximation is not exact
- The closer x_1 is to x_2 the more accurate the approximation

Taylor series

- We can quantify the error in a finite difference expression by considering Taylor series

- A Taylor series approximates a function at $x + \Delta x$ in terms of $f(x)$ and its derivatives (evaluated at x)

$$f(x + \Delta x) = f(x) + \frac{\partial f}{\partial x}\Delta x + \frac{\partial^2 f}{\partial x^2}\frac{(\Delta x)^2}{2} + \cdots + \frac{\partial^n f}{\partial x^n}\frac{(\Delta x)^n}{n!}$$

Taylor series

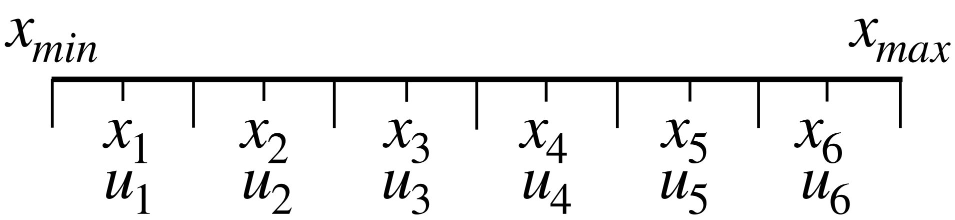
$$f(x+\Delta x) = f(x) + \frac{\partial f}{\partial x}\Delta x + \frac{\partial^2 f}{\partial x^2}\frac{(\Delta x)^2}{2} + \cdots + \frac{\partial^n f}{\partial x^n}\frac{(\Delta x)^n}{n!}$$

Apply to the discrete representation of $u(x)$

$$u_{i+1} = u_i + \frac{\partial u_i}{\partial x}\Delta x + \frac{\partial^2 u_i}{\partial x^2}\frac{(\Delta x)^2}{2} + \frac{\partial^3 u_i}{\partial x^3}\frac{(\Delta x)^3}{6} + \cdots$$

Rearrange to put $\dfrac{\partial u_i}{\partial x}$ on the left hand side

$$\frac{\partial u_i}{\partial x} = \frac{u_{i+1} - u_i}{\Delta x} - \left[\frac{\partial^2 u_i}{\partial x^2}\frac{\Delta x}{2} + \frac{\partial^3 u_i}{\partial x^3}\frac{(\Delta x)^2}{6} + \cdots\right]$$



Taylor series

Finite difference representation

Truncation error

$$\boxed{\frac{\partial u_i}{\partial x} = \frac{u_{i+1} - u_i}{\Delta x}} - \boxed{\left[\frac{\partial^2 u_i}{\partial x^2}\frac{\Delta x}{2} + \frac{\partial^3 u_i}{\partial x^3}\frac{(\Delta x)^2}{6} + \cdots\right]}$$

$$\frac{\partial u_i}{\partial x} = \frac{u_{i+1} - u_i}{\Delta x} \boxed{+ \mathcal{O}(\Delta x)}$$

Leading order term in the truncation error is $\mathcal{O}(\Delta x)$

Truncation error

- The truncation error is $\mathcal{O}(\Delta x)$ so this finite difference is said to be first order accurate
- The truncation error is reduced as Δx gets smaller
- The truncation error is inherent to the numerical scheme
- There is also round off error due to floating point precision (more on this later in the module)

Other finite difference stencils

- There is more than one way to calculate a gradient e.g.
  - `dudx = ( u(i+1) - u(i)   )/dx`
  - `dudx = ( u(i)   - u(i-1))/dx`
  - `dudx = ( u(i+1) - u(i-1))/(2*dx)`
- The finite difference representation affects the accuracy
- The first two expressions are first order accurate but what about the last one?

Other finite difference stencils

```
dudx = ( u(i+1) - u(i-1))/(2*dx)
```

$$u_{i+1} = u_i + \frac{\partial u_i}{\partial x}\Delta x + \frac{\partial^2 u_i}{\partial x^2}\frac{(\Delta x)^2}{2} + \frac{\partial^3 u_i}{\partial x^3}\frac{(\Delta x)^3}{6} + \cdots$$

$$u_{i-1} = u_i - \frac{\partial u_i}{\partial x}\Delta x + \frac{\partial^2 u_i}{\partial x^2}\frac{(\Delta x)^2}{2} - \frac{\partial^3 u_i}{\partial x^3}\frac{(\Delta x)^3}{6} + \cdots$$

Subtract and re-arrange

$$\frac{\partial u}{\partial x} = \frac{u_{i+1} - u_{i-1}}{2\Delta x} + \mathcal{O}\left(\Delta x\right)^2$$

This finite difference is second order accurate. The truncation error approaches zero more rapidly as Δx approaches zero

Unit Summary

- Finite differences can be used to calculate an approximation to a spatial derivative
- We can use Taylor series to quantify the error in the approximation (the **truncation error**)
- The size of the truncation error depends on the spacing between grid points (**spatial resolution**)
- The leading term in the truncation error depends on the finite difference being used and affects the accuracy